

## PACK WITH LOVE CLUB

## From Data to Impact · Course Map

*What each lesson builds — business, math, statistics — and where it sits in the revised AP Statistics course · 2026–27 · companion to the 24-Lesson Curriculum*

**Reading the AP column.** Topic numbers are Unit.Topic (1.1–1.13; 2.1–2.12). Skills: 1.A determine an investigative question · 2.A identify information · 2.B justify an ethical collection method · 3.A construct representations · 3.B calculate summary statistics · 3.C estimate counts, percentages, probabilities · 4.A describe and compare · 4.B justify a claim · 4.C describe distributions · 4.D interpret results. Practices: P1 Formulate Questions · P2 Collect Data · P3 Analyze Data · P4 Interpret Results. Concept level only; inference (Units 3–5) is out of scope by design.

| #  | Date  | Lesson                                                   | Practice                   | Business capability                                                                                                                                                              | Math capability (modeling & quantitative reasoning)                                                           | Statistical reasoning                                                                                                                                           | AP (topics · skills)                                             |
|----|-------|----------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1  | 9/13  | What Is a Business? (Prof. Sun)                          | —                          | Four-box business model; recognizing risk; who decides on funds                                                                                                                  | Revenue – Cost = Profit as an equation; price × units; estimation (MP4, MP2)                                  | Seed: how would you find out?                                                                                                                                   | seed of 1.A                                                      |
| 2  | 9/20  | The Birth of a Biscuit (Joyce Xie)                       | —                          | The full product chain on one real case; the four research questions; no perfect supplier (quality–speed–price); MOQ, sampling, defects, lead time; four risks, four plans       | Order of magnitude; cost structure; cost/time/risk as three dimensions of one decision                        | Seed: small-batch trial and iteration (the Event I → II idea)                                                                                                   | —                                                                |
| 3  | 9/27  | Choose the Product, Find the Source (Tony Yao)           | —                          | Three entry points; product screening; reading a listing; listed vs. quoted; landed-cost estimate; inquiry drafting; product decision                                            | First landed-cost model — multi-step percent, unit rate (7.RP); a source and an uncertainty per number (MP6)  | Source-reliability habits (P2/P4, not yet named)                                                                                                                | habits for 2.A / 4.D                                             |
| 4  | 10/4  | Pose the Statistical Question — Who Are We Asking?       | <b>Question</b>            | Turning a business unknown into a researchable question                                                                                                                          | —                                                                                                             | Investigative question, three components, fixed; population/sample/census; observational units; variables; parameter vs. statistic named; prediction in context | 1.1 (1.1.A, 1.1.B taught) · 1.2 · 1.10.A–B · 1.10.D.4 · 1.A, 2.A |
| 5  | 10/11 | Bias, Two Ways — Whom We Ask and How We Ask              | <b>Collect</b>             | Asking a school or studio for permission yourself (the Bring It guides); reporting where you stand                                                                               | Fractions and quotas                                                                                          | SRS, stratified, convenience; bias = systematic error; voluntary response, undercoverage; wording bias; honest generalization; Survey v1                        | 1.11 · 1.12 · 1.10.E · 2.A, 2.B                                  |
| 6  | 10/18 | Survey Launch Lab → Survey Week 10/18–25                 | <b>Collect</b>             | Field discipline; ethical customer contact                                                                                                                                       | Coding scheme                                                                                                 | Simulation of friends-only bias; quota cards; pilot; interviewer effect; nonresponse as data                                                                    | 2.3 · 1.12 · 1.10.E · 2.A, 2.B, 3.C                              |
| 7  | 10/25 | Raw Data → Percentages & Averages                        | <b>Analyze</b>             | Reading a market                                                                                                                                                                 | Fraction → percent by hand; mean, median, range; five-number summary (ext.)                                   | Distributions; parameter vs. statistic; sampling variability by school; $1/\sqrt{n}$                                                                            | 1.2.A · 1.3 · 1.5–1.9 · 2.12 · 3.A, 3.B, 4.A, 4.C                |
| 8  | 11/1  | Graphs, Evidence & the Price Signal                      | <b>Analyze → Interpret</b> | Market brief; segments; price range; skepticism toward AI                                                                                                                        | Row percentages; reading a truncated axis                                                                     | Joint/marginal/conditional; association; CER; predictions opened                                                                                                | 1.4 · 1.9 · 2.1–2.2 · 3.A, 3.B, 4.A, 4.B, 4.D                    |
| 9  | 11/8  | Samples in Hand — the Mathematics of a Weighted Decision | —                          | Supplier evaluation: sample QC at home; quoted vs. listed; the Jan 10 knock-out; China and US wholesale on equal footing; weighted matrix (price as landed cost); recommendation | Weighted average $\Sigma(w \times s) \div 100$ and why weights total 100; the flip-point inequality (7.EE.4)  | Evidence-cited scoring                                                                                                                                          | 3.B · 7.EE.4                                                     |
| —  | 11/15 | <i>no class</i>                                          |                            |                                                                                                                                                                                  |                                                                                                               |                                                                                                                                                                 |                                                                  |
| 10 | 11/22 | Cost per Unit, Quantity & the Order (order ≈ 11/23)      | —                          | Unit economics; quantity with buffer; budget ceiling; go/no-go; confirming the path                                                                                              | Cost model fixed + variable × qty; unit rate (7.RP.1); percent buffer (7.RP.3); budget as inequality (7.EE.4) | Optimism haircut                                                                                                                                                | —                                                                |
| —  | 11/29 | <i>Thanksgiving</i>                                      |                            |                                                                                                                                                                                  |                                                                                                               |                                                                                                                                                                 |                                                                  |
| 11 | 12/6  | Landed Cost — the Real Numbers, and What We Didn't Buy   | —                          | Import math on the chosen path; domestic vs. import; tariff risk; MOQ trap in hindsight                                                                                          | Multi-step percent with real numbers; percent error; sensitivity one input at a time (MP2)                    | —                                                                                                                                                               | —                                                                |

| #  | Date         | Lesson                                                                  | Practice                                  | Business capability                                                                          | Math capability (modeling & quantitative reasoning)                          | Statistical reasoning                                                                                                                                              | AP (topics · skills)                                         |
|----|--------------|-------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| 12 | 12/13        | Pricing Scenarios — the Data Sets the Price                             | <b>Interpret (cycle 1 closes)</b>         | Pricing strategy; break-even; sell-through                                                   | Scenario table; break-even = fixed ÷ (price – unit cost) (MP4)               | Discounting stated intent; assumptions named                                                                                                                       | 4.D habit                                                    |
| —  | 12/20, 12/27 | <i>winter break</i>                                                     |                                           |                                                                                              |                                                                              |                                                                                                                                                                    |                                                              |
| 13 | 1/3          | Mission-Based Marketing                                                 | <b>Question → Collect (cycle 2 opens)</b> | Value proposition; Charter \$8 messaging; message test                                       | —                                                                            | Experiment: units, treatments, response, random assignment by coin flip; causal scope                                                                              | 1.1.B applied · 1.10.C · 1.13.A, D · 1.A, 2.A, 2.B           |
| 14 | 1/10         | The Boxes Are Open — Arrival Inspection, and What a Sample Can Tell You | Collect → Analyze                         | Receiving; defect definition; acceptable rate; claim path; risk register (business 15 min)   | Percent of a quantity; count reconciliation                                  | Random sample of units; <b>simulation of a 3% batch — fifteen draws of 40 into a dotplot</b> ; sampling distribution of a proportion                               | 1.2 · 1.11 · 2.3 · 2.12 · 2.A, 3.A, 3.C, 4.C                 |
| 15 | 1/17         | Forecast Demand & Choose the Market                                     | <b>Question → Analyze</b>                 | Demand forecast; <b>one market for both events</b> ; audience-fit gap; adult mini-survey     | Multiplicative model traffic × stop × buy; expected revenue = price × buyers | Random process; law of large numbers; expected value; to whom conclusions apply                                                                                    | 1.1.B applied · 1.10.E · 2.3 · 2.4 · 2.9 · 1.A, 2.A, 3.C     |
| 16 | 1/24         | Sales System, Event Day Kit & Two Ways to Run an Experiment             | <b>Collect (design)</b>                   | Point-of-sale system; cash controls; roles; market outreach                                  | Conversion = buyers ÷ stopped; revenue per buyer; units per hour             | <b>Design 1</b> whole-event (no causal claim) vs. <b>Design 2</b> randomized alternating slots (causal claim, scoped); confounding; replication                    | 1.10.C–D · 1.13.A, D · 2.A, 2.B                              |
| 17 | 1/31         | Home Practice Sale — and Turning the Forecast into a Distribution       | Collect → Analyze                         | Selling at home; reconciliation; Kit v2; inventory split (business 20 min)                   | Units × price = cash; a percentile-based stock decision                      | <b>Simulation of fifteen Saturdays into a distribution</b> ; expected value with variability                                                                       | 2.3 · 2.9 · 2.12 · 2.A, 3.A, 3.C, 4.C                        |
| 18 | Sat 2/6      | Sales Event I (same market)                                             | <b>Collect</b>                            | Running a real sale; cash rule; adult mini-survey at the table                               | Real-time tally; running conversion                                          | Observations on defined variables; the baseline                                                                                                                    | 1.2 · 2.A                                                    |
| 19 | 2/7          | Analyze & Adjust — Which Variables May Be Randomized?                   | <b>Analyze → Interpret → Question</b>     | One change, committed in writing, with its design                                            | Percent error by input; percent change; numeric prediction                   | Stated vs. revealed (students and adults); slot-switchable vs. whole-event-only adjustments; <b>price cannot ethically be randomized</b> ; Event II question fixed | 1.1.B student-generated · 1.13.A · 2.12 · 1.A, 3.B, 4.B, 4.D |
| —  | 2/14         | <i>reserve Sunday</i>                                                   |                                           |                                                                                              |                                                                              |                                                                                                                                                                    |                                                              |
| 20 | Sat 2/20     | Sales Event II / Campaign Close (same market)                           | <b>Collect</b>                            | Executing the adjustment on the coin-flip slot schedule; unsold accounting                   | Cumulative totals; comparison by slot                                        | The randomized schedule executed; replication                                                                                                                      | 1.13.A · 2.A                                                 |
| 21 | 2/21         | Post-Sale Statistical Analysis                                          | <b>Analyze → Interpret</b>                | Campaign review by segment; honest attribution; an inconclusive result reported as a finding | Percent change; ratio stated/revealed; hand-verification (MP6)               | <b>A/B two-way table and its causal scope</b> ; whole-day comparison and its limits; confounding; verify AI                                                        | 2.1–2.2 · 1.13.D · 1.10.D.5 · 3.A, 3.B, 4.A, 4.B, 4.D        |
| 22 | 2/28         | Profit, Net Fundraising & Impact Math                                   | —                                         | Net vs. gross; inventory cost; impact accounting; explaining to a parent                     | Revenue – ordered cost – expenses = net; margin; net as % of \$5,300         | Percent of a whole, honestly labeled                                                                                                                               | —                                                            |
| 23 | 3/7          | Business Review                                                         | <b>Interpret (cycle 2 closes)</b>         | Post-mortem; decision vs. outcome; Year 2 Brief                                              | 2×2 reasoning under uncertainty                                              | Evidence synthesis; the question in reverse                                                                                                                        | 1.1.B reflected · 4.B, 4.D                                   |
| 24 | Sun 3/14     | Final Impact Showcase                                                   | All four                                  | Public accountability; portfolio                                                             | One verified number per beat                                                 | Data storytelling                                                                                                                                                  | 1.A–4.D · P1–P4                                              |

## Coverage summary

- **Unit 1 (Exploring One-Variable Data and Collecting Data):** all 13 topics at the concept level — 1.1–1.2 (L4, L7), 1.3–1.9 (L7–L8), 1.10 (L4, L5, L16), 1.11 (L5, L14), 1.12 (L5–L6), 1.13 (L13, L16–L21).
- **Unit 2 (Probability, Random Variables, and Probability Distributions):** 2.1–2.2 (L8, L21), 2.3 (L6, L14, L15, L17), 2.4 (L15), 2.9 (L15, L17), 2.12 by simulation and intuition (L7, L14, L17, L19).
- **Skills:** 1.A, 2.A, 2.B, 3.A, 3.B, 3.C, 4.A, 4.B, 4.C, 4.D — the ten non-inference skills, and the two free-response questions (FRQ 1: Practices 1–2; FRQ 2: Practices 3–4) that assess them.

• **Lesson types:** 9 led by statistics (L4–L8, L14–L17 in their second halves, L19, L21) · 4 by business math (L10–L12, L22) · 7 by business and operations (L1–L3, L9, L13 first half, L14 and L17 first halves) · 2 fieldwork (L18, L20) · 2 synthesis (L23, L24).